

AMINOGLYCOSIDES

- Powerful class of antibiotics, they are the BIG GUNS
- When nothing else works, pull out the aminoglycosides
- Think the following: **A MEAN OLD MYCIN**

A) They are antibiotics used to treat A MEAN OLD INFECTION that is RESISTANT SERIOUS LIFETHREATENING RESISTANT GRAM NEGATIVE

- Sinusitis? No. It's not a mean old infection.
- TB? Yes.
- Otitis Media? No.
- Bladder infection? No.
- Fulminant pyelonephritis? Yes.
- Septic shock? Yes.
- Burn wounds over 85% body? Yes.
- Viral pharyngitis? No.
- Strep throat? No.

B) "Mycin" → ALL aminoglycosides end in -mycin

- Not all drugs that end in mycin are aminoglycosides but MOST are
- There are three mycins that are not aminoglycosides
 - Arithromycin
 - Zithromycin
 - Clarithromycin
 - If it ends in MYCIN = Aminoglycoside
 - If it ends in THROMYCIN = It is NOT an aminoglycoside, which you can use for regular type infections

C) TWO TOXIC EFFECTS OF A MEAN OLD MYCIN / AMINOGLYCOSIDES

- Think MICE → think EARS → **ototoxicity**
 - Hearing*
 - Ringing/Tinnitus
 - Vertigo (equilibrium)
 - Dizziness (equilibrium)
- Human ear is shaped like a KIDNEY → **nephrotoxicity**
 - Creatinine is THE best indicator of kidney function (do not go for U/O, BUN)
 - 24 hr creatinine clearance is BETTER than serum creatinine
 - Serum creatinine comes next over anything else
- Think number 8 drawn inside the EAR
 - Toxic to Cranial Number 8 (ear nerve)
 - Administer them q8h
- Route: IM or IV
- Do not give them PO because they are not **absorbed EXCEPT in**
 - **Hepatic Encephalopathy**
 - Oral mean old mycins will go in your gut and kill gram negative bacteria which produce ammonia
 - This will help get rid of high ammonia levels
 - At the same time, the drug will NOT go to the liver (since it's not absorbed) which is GOOD in liver failure
 - **Preop Bowel Surgery**
 - Clean out the gut

- NO OTOXOCITY OR NEPHROTOXICITY in both cases because not absorbed
- In both cases → sterilized bowel
- Therefore, ORAL MEAN OLD MYCINS are ORAL BOWEL STERILIZERS
- **NEOMYCIN and CANOMYCIN are the top 2 oral bowel sterilizers**
- **Who CAN sterilize my bowel? NEO CAN.**

AMINOGLYCOSIDE = A MEAN OLD MYCIN

“TAP”

TROUGH: Drug at its lowest

ADMINISTER

PEAK: Drug at its highest

Reason for trough levels?

- Narrow therapeutic window
- Small difference between what works and what kills
- Draw TAPS

Example: LASIX

Smallest dose seen? 5mg or 10mg.

Largest dose seen? 120 mg

Therefore, WIDE range. No TAPS.

Example: DIGOXIN

Lowest: 0.125

Largest: 0.25

Therefore, NARROW range. TAPS, yes.

A MEAN OLD MYCINS = TAPS DRAWN!

When should TAPS be drawn? What matters? **THE ROUTE.**

TAPS

	TROUGH	PEAK
SL	30 mins before next dose	5-10 minutes after drug dissolved
IV	30 mins before next dose	15-30 minutes after drug is finished, not when you hang it IE: 100 ml at 200 ml/hr. It will be finished in 30 mins. You hang it at 1000hr, finishes at 1030hr. Draw peak from 1045-1100hr. If you get two correct values in range; play price is right. Choose the one that is the highest without going over. Therefore, 1100hr.
IM	30 mins before next dose	30-60 minutes after injected. Choose 60 minutes, if both options given.
SC	30 mins before next dose	SEE diabetes lecture. Only SC peaks they talk about are insulins.
PO	30 mins before next dose	Forget about it because they don't test it. Too variable.

CALCIUM CHANNEL BLOCKERS

- CCB are like VALIUM for your heart.
- What does VALIUM do for your body? Calms you down.
- CCBs calm the heart down.
 - If your heart is tachycardic, can it stand a little relaxant → YES CCB
 - If in shock, does it need to relax? → NO CCB
 - Heart block → NO CCB
 - Tachyarrhythmias → YES CCB
 - Heart attack → YES CCB

A. CCBs are negative inotropic, negative chronotropic, and negative dromotropic

- It's like VALIUM for your heart. Don't freak out with the fancy words. They relax your heart.

Positive ino/chrono/dromo	Negative ino/chrono/dromo
Cardiac stimulants Strengthen, speed up, make it work harder	Cardiac depressant Slow down and depress the heart

B. What do they treat? AAA!

- **Antihypertensives:** They relax your heart and blood vessels
- **Antianginals:** Relax the heart so it uses less oxygen; decreasing oxygen demand
- **Anti atrial-arrhythmias:** ONLY atrial arrhythmias NOT ventricular. Treats atrial flutter, atrial fib, paroxysmal atrial tachycardia... anything ATRIAL!
 - **There's a trick:** supraventricular tachycardia. Would CCB treat this? YES. Why? "SUPRA" means above; therefore, ABOVE THE VENTRICLES = ATRIAL.
 - SVT = ATRIAL
- **Side Effects**
 - Hypotension: Vasodilation
 - Headache: Vasodilation in the brain
 - Headache is usually always in SATA
- **Always measure blood pressure due to risk for hypotension**
- **Parameters: Hold CCB if SYSTOLIC < 100 mmHg**

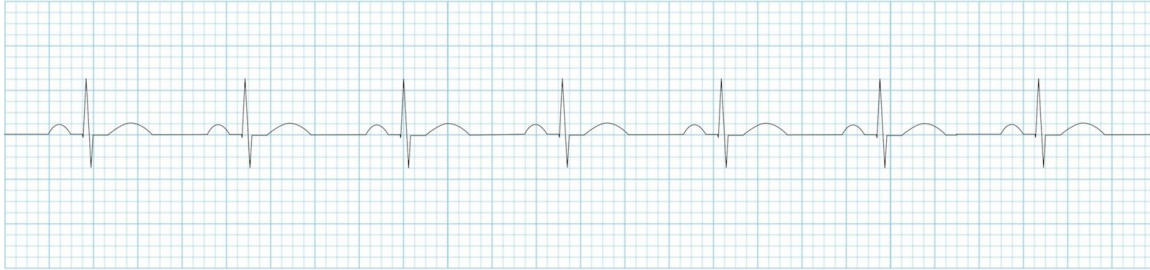
C. Names of CCB

- **-dipine**
- Has to be Dlpine not PINE
- Two others:
 - Verapamil
 - Cardizem
- Cardizem can be given continuous IV
 - If SBP < 90 mmHg, slow and titrate the drip to keep SBP > 100 mmHg

CARDIAC ARRHYTHMIAS

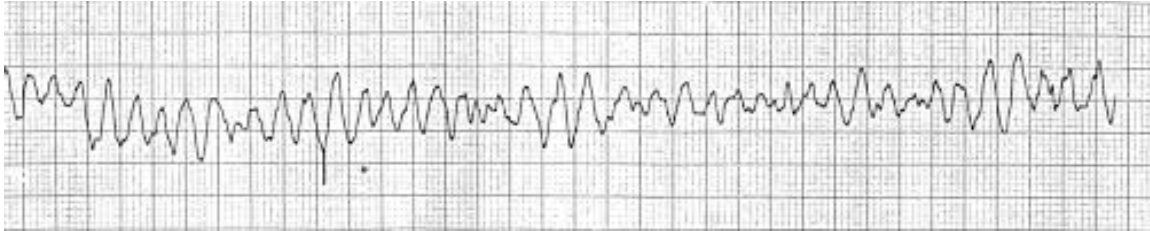
Four rhythm strip tracings; know by SITE.

1. Normal Sinus Rhythm



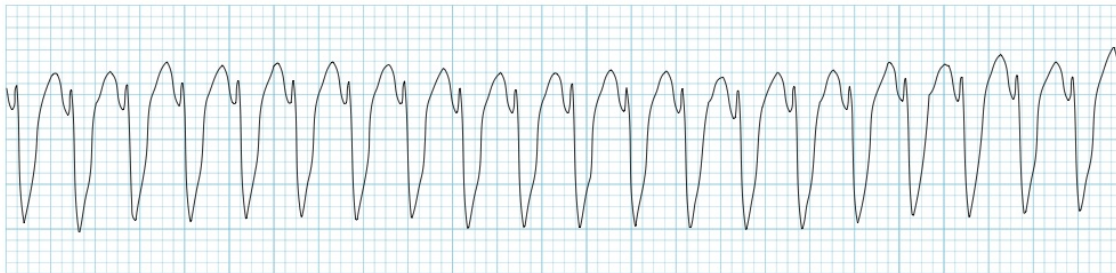
- There's a P wave, before every QRS, which is followed by a T wave
- Peaks of P waves are equally distant from one another → NOT a sinus arrhythmia

2. Ventricular Fibrillation



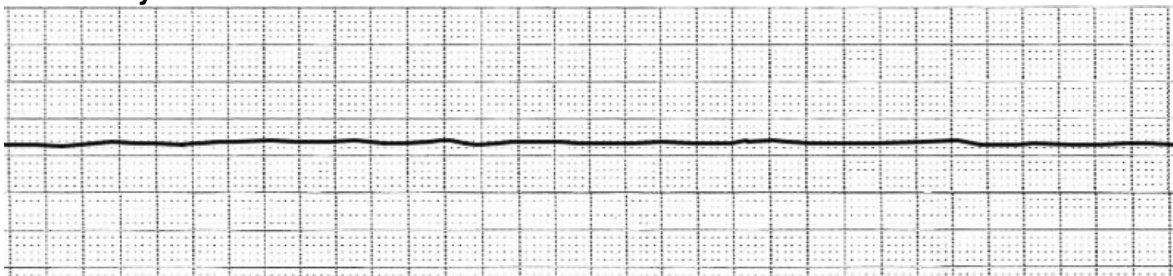
- Chaotic squiggly line
- Is there a pattern? NO.

3. Ventricular Tachycardia



- Sharp peaks and jags
- Is there a pattern? YES.

4. Asystole



TERMINOLOGY

- **QRS: VENTRICULAR**
- **P WAVE: ATRIAL**

6 RHYTHMS MOST TESTED ON NCLEX

1. A lack of QRS's = no QRS → **asystole**
2. Saw tooth = flutter → **atrial flutter**
3. Chaotic = fibrillation → **atrial fibrillation with P wave**
4. Chaotic = fibrillation → **ventricular fibrillation with QRS**
5. Bizarre = tachycardia → **ventricular tachycardia**
6. Periodic widened QRS → **PVC; one snapshot of tachycardia**
 - a. PVC's are LOW priority unless...
 - i. If there are more than 6 PVC's in a minute
 - ii. If there are more than 6 PVC's in a row
 - iii. If the PVC falls on the T wave of the previous beat
 - b. If one of the three are true, you elevate priority of PVC client to MODERATE
 - c. PVC's never reach HIGH priority level

LETHAL ARRHYTHMIAS: Kill you in 8 minutes or less, HIGH PRIORITY.

1. Asystole
2. Ventricular Fibrillation

No cardiac output → no brain perfusion → dead in 8 minutes

POTENTIALLY LIFE THREATENING

1. Ventricular Tachycardia

There IS cardiac output

*When the rhythm changes, doctor will ask... DO YOU GET A PULSE WITH THAT?
If there is a pulse = THERE IS CARDIAC OUTPUT*

TREATMENT

1. **VENTRICULAR TACH and PVCs:** Amiodarone or lidocaine.
2. **ATRIAL ARRHYTHMIAS/SUPRAVENTRICULAR TACHYCARDIA:** ABCD's
 - **Adenosine/ADENOCARD → push in less than 8 seconds → FAST IV PUSH**
 - IV PUSH: When you don't know, you go slow BUT THIS IS ONE YOU HAVE TO KNOW!!! Use BIG vein.
 - Can go into asystole for 30 seconds since it's such a fast push BUT they should come back!!!
 - **Beta-blockers → "-lol"**
 - Negative ino/chrono/drono = they are like valium for your heart
 - Treat AAA and AA
 - Therefore SE → Hypotension & headache just like CCB
 - **Calcium Channel Blockers → like VALIUM for your heart; same as Beta Blockers**
 - **Digitalis → digoxin, lanoxin KNOW NAMES!****
3. **VFIB:** For VFIB you DFIB
 - Shock them!
4. **ASYSTOLE:** Epinephrine and atropine in that order

CHEST TUBES

- Re-establishes negative pressure in pleural space so that the lung expands when the chest wall moves
- Negative pressure is good in pleural space = makes things stick together (visceral and parietal layer)
- Positive pressure pulls things apart = more work; less air; because of positive pressure

A. When you get a question, look for reason it was placed

- **Pneumothorax:** chest tube to remove air
- **Hemothorax:** chest tube removes blood
- **Pneumo-hemo thorax:** chest tube removes both air and blood

Examples:

CT for HEMOTHORAX – what would you report?

1. No bubbling
2. CT drained 800 ml in first 10 hours
3. **CT is not draining → b/c it is not doing what it's supposed to do**
4. CT is intermittently bubbling

CT for PNEUMOTHORAX – what would you report?

1 or 2.

1. It's supposed to bubble
2. It's not supposed to drain!

B. Location of tube placement → A for A, B for B!

1. **Apical:** chest tube is way up high → draining air because air rises

2. **Basilar:** chest tube is at bottom of lungs → draining blood because it is subject to gravity

Examples:

Your apical CT is draining 300ml/hr → BAD!

Your basilar CT is **NOT bubbling → GOOD!**

Hemo → Basilar

Pneumo → Apical

Pneumo/Hemo → BOTH

How many CT and where would they be placed for unilateral pneumohemothorax?

Two. Apical for pneumo. Basilar for hemo.

How many CT and where would they be placed for bilateral pneumothorax?

Two. Both apical.

How many CT and where would they be placed for post op chest sx?

Two. Apical and basilar on side of sx.

Always assume UNILATERALITY.

Only time you're taking care of BILATERAL is when they say.

How many CT and where would you place them for post op right pneumonectomy?

REMOVAL OF LUNG so NONE!!!

Only for lobectomy, wedge resections, etc. but NOT PNEUMONECTOMY.

TROUBLESHOOTING

- **Closed chest drainage systems** (pneumovac, pleuravac, emerson – plastic containers w/ tube connected)
 1. If you knock it over → set it back up and they need to take deep breaths → not a medical emergency
 2. If device breaks → positive pressure can get into pleural space
 - a. CLAMP it so nothing gets in
 - b. CUT the tube away from broken device
 - c. SUBMERGE end of tube into sterile water
 - d. UNCLAMP it b/c re-established water seal
 - e. That is the order (alphabetical)!
 - f. NOTE: best question is different than a first question
 - i. BEST = what's the one thing you could do if you only had one choice?
SUBMERGE.
 - ii. FIRST = what would you do first? **CLAMP.**
 3. If the CT gets pulled out
 - a. FIRST = Take a gloved hand and cover the hole
 - b. BEST = Cover in vaseline gauze
 4. Bubbling → sometimes it's good sometimes its bad...
 - a. Ask yourself two questions → Where is it bubbling? When is it bubbling?
 - b. **INTERMITTENT WATERSEAL** = always good, document it.
 - c. **CONTINUOUS WATERSEAL** = there is a leak, this is bad. Find it and tape it until it stops leaking.
 - d. **INTERMITTENT SUCTION CONTROL CHAMBER** = bad, this is not high enough. Suction is too low. Go to the wall and turn it up until continuous bubbling.
 - e. **CONTINUOUS SUCTION CONTROL CHAMBER** = good. Document it.
 - f. **TWO GOOD AND TWO BAD SCENARIOS:**
 - i. Just remember one situation, and the other is opposite.
 - ii. Continuous bubbling in water seal is like a bottle of pop. If it was continuously bubbling, would you buy it? NO. There's a leak. Just like a CT.

RULES FOR CLAMPING TUBES

- Never clamp CT for more than 15 seconds, without a doctor's order.
- Use rubber tipped double clamps
 - The teeth of the clamps are rubber tipped to prevent puncture of tube
 - Double clamps just because...

CONGENITAL HEART DEFECTS

- Every congenital heart defect is either trouble or no trouble. It either causes a lot of problems or it's not big deal at all. There is no in between.
- Memorize ONE word → **TRouBLE**
- Nurses role in defects → teach about implications, NOT diagnosis.

TROUBLE DEFECT	NO TROUBLE DEFECT
You need surgery, now, in order to live.	You do not need surgery, you can possibly have it, years later, if it causes trouble, but usually does not cause trouble.
Growth is slow and delayed.	Growth is normal.
Life expectancy short.	Normal life expectancy.
Parents → stress, financial, grief, etc.	Parents → average life.
D/C with apnea monitor.	No apnea monitor.
Pediatric cardiologist.	Pediatrician/NP.
Weeks in hospital after birth.	24-36 hours after birth.
Shunts RIGHT TO LEFT because that's the way TROUBLE is spelled.	Shunts blood LEFT TO RIGHT because it's not the way TROUBLE is spelled. It is no trouble.
TRouBLE → RIGHT TO LEFT is BLUE → Cyanotic; because TROUBLE is spelled that way.	Acyanotic (pink).
They will all have murmurs because of SHUNT.	
They will all have ECG done to find out why.	

- There are 47 heart defects.
- All congenital heart defects that start with T are trouble. TRouBLE.

Examples:

Ventricular septal defect → no trouble

Tetralogy of Fallor → TROUBLE

Patent foramen ovale → no trouble

Truncus arteriosus → TROUBLE

Transposition of great vessels → TROUBLE

Etc.

- **Only ONE exception**
 - LEFT VENTRICULAR HYPOPLASTIC SYNDROME
 - Boards will not bring this up

FOUR DEFECTS OF TETRALLOGY OF FALLOR:

“VarieD PictureS Of A Ranch”

1. VD → ventricular defect
2. PS → pulmonary stenosis
3. OA → overriding aorta
4. RH → right hypertrophy

Don't memorize what they are. Just memorize the names.